Requirements Gathering

Approach: The objective is to analyze how crises affect ship traffic at the main maritime chokepoints and how long recovery takes. The scope is the PortWatch data from 2019 to the present, over 4 weeks. We gathered the requirements by brainstorming as a team and by exploring the PortWatch data. We recorded everything in one shared requirements list, with an ID for each item (FR = functional, NFR = non-functional) so we can trace it to the SQL, the notebooks, and the dashboard.

1. Stakeholder Analysis

Stakeholder	**Need**	**Influence / Interest**
Executive manager	Which routes are at risk and how long a crisis lasts, to plan supply risk	High / High
Shipping and logistics manager	How much traffic dropped, where it moved, and how many days to recover	Medium / High

2. User Stories & Use Cases

User stories

- As an executive manager, I want to see which chokepoints are most vulnerable, so I can plan for supply risks.
- As a logistics manager, I want to see how much traffic fell during a crisis compared with normal, so I can understand its impact.
- As a logistics manager, I want to see how many days it took to return to normal, so I can estimate how long a future crisis may last.
- As a logistics manager, I want to see whether traffic moved to alternative routes, so I can consider other options.

Use case: View the impact of a crisis

- **Actor:** logistics manager or executive manager.

Steps:

1) Open the dashboard.
2) Select a chokepoint, an event, and a ship type.

3) The dashboard shows traffic and estimated cargo compared with the baseline.
4) The user drills down to the recovery timeline.

- **Result:** the user sees the size of the drop and the recovery time.

3. Functional Requirements (with MoSCoW priority)

ID	Requirement	Priority
FR1	Load the PortWatch data into Bronze, clean it in Silver, and build the Gold model	Must
FR2	Show traffic and estimated cargo volume by chokepoint over time	Must
FR3	Compare each crisis with a pre-event baseline (% drop)	Must
FR4	Calculate recovery time in days	Must
FR5	Filters by chokepoint, ship type, date, and event, with drill-down	Must
FR6	Show traffic changes on alternative routes	Should
FR7	Rank chokepoints by vulnerability	Should
FR8	Link disruption events to the affected chokepoints	Should

Won't have: prices, costs, tariffs, forecasting, and port-level analysis.

4. Non-functional Requirements

ID	Area	Requirement
NFR1	Performance	The dashboard responds quickly when filters change
NFR2	Security	Only public data, no personal data, and we follow the PortWatch license and citation
NFR3	Usability	A simple, clear layout that non-technical managers can read, with labeled charts
NFR4	Reliability	Quality checks in each layer, and the dashboard numbers match the SQL results

Constraints: the cargo values are preliminary estimates, there is no price or cost data, we use a fixed snapshot of the data with the download date recorded, and we have 4 weeks.

Verification and Changes

- **Validation:** we will show the requirements list and a dashboard wireframe to the lecturer and record the approval before we build.
- **Change control:** any change to a requirement goes into a change log (date, what changed, why, who approved).